PS-SSR: Methods Beating Pure SSR on 6Q Benchmark

Qwen3-8B (bf16) · 66 clusters · ∼12K posts · 6 original survey questions · Mean Jensen-Shannon divergence

Baseline: Direct SSR = **0.0277**. Lead method: **SSR + LLM-dist (w=0.5)** = **0.0257**, −7.2%.

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1. Methods

M0 Pure SSR (baseline) — Embedding-based. Embed cluster posts and the 5 anchor options; per-cluster distribution is softmax of anchor-to-post similarity. Aggregate uniformly over top-K clusters. No LLM at inference.

M1 LLM-dist — For each cluster, feed the LLM a sample of posts and the 5 anchors in a single prompt; the LLM outputs a probability over the anchors directly (not per-post voting). Aggregate across clusters.

M2 SSR + LLM-dist ensemble ★ — Linear mix of M0 and M1: $P = w \cdot P_{\text{SSR}} + (1-w) \cdot P_{\text{LLM-dist}}$. Best at $w = 0.5$.

M3 Multi-layer persona-steered generation — Extract a per-cluster persona direction from mean hidden states at layers $\{16, 20, 24\}$. At generation, add all three steering vectors simultaneously, sample n responses per cluster, then run SSR on the steered outputs.

M4 SSR + ML-steer ensemble — Linear mix of M0 and M3: $P = w \cdot P_{\text{SSR}} + (1-w) \cdot P_{\text{ML-steer}}$. Best at $w = 0.5$–0.7.

M5 SSR knob tuning — Pure SSR with swept posts-per-cluster n or top-K. No LLM.

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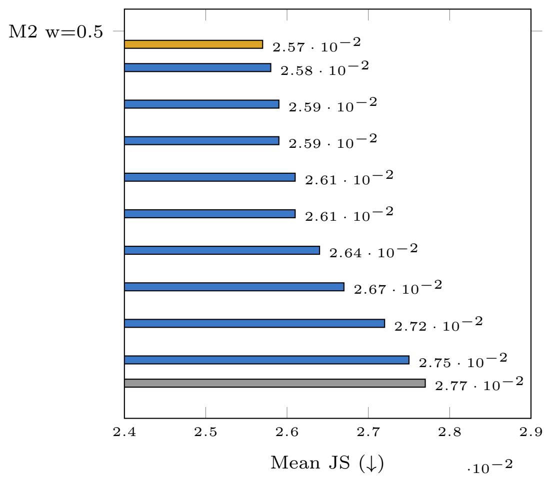
2. Master Leaderboard

Table 1: Mean JS on 6Q (↓ better)

#	Method	Mean JS	Δ vs M0
1	M2: SSR + LLM-dist, w=0.5	**0.0257**	**−7.2%**
2	M2: SSR + LLM-dist, w=0.6	0.0258	−6.9%
3	M2: SSR + LLM-dist, w=0.7	0.0259	−6.5%
4	M1: LLM-dist top-10	0.0259	−6.5%
5	M2: SSR + LLM-dist, w=0.8	0.0261	−5.8%
6	M4: SSR + ML-steer, w=0.5	0.0261	−5.4%
7	M4: SSR + ML-steer, w=0.7	0.0264	−4.3%
8	M4: SSR + ML-steer, w=0.9	0.0267	−3.3%
9	M5: SSR top-5 clusters	0.0272	−1.8%
10	M5: SSR n=50 posts	0.0275	−0.7%
—	*M0: Pure SSR baseline*	*0.0277*	—

Figure 1: Mean JS (shorter = better)

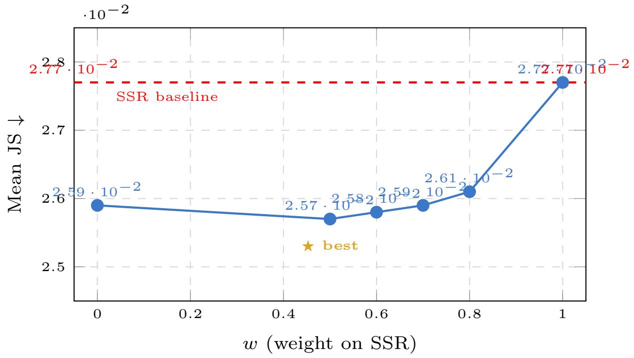


3. Per-question JS (top methods)

Method	Q1 `8.0/3`	Q2 `8.0/4`	Q3 `9.0/3`	Q4 `9.0/4`	Q5 `9.0/5`	Q6 `11.0/3`
M2 SSR+LLMdist w=0.5	0.0098	0.0483	0.0330	0.0155	0.0339	0.0139
M2 SSR+LLMdist w=0.6	0.0103	0.0476	0.0303	0.0157	0.0362	0.0146
M2 SSR+LLMdist w=0.7	0.0109	0.0470	0.0275	0.0159	0.0386	0.0154
M1 LLM-dist top-10	0.0027	0.0493	0.0466	0.0195	0.0310	0.0062
M2 SSR+LLMdist w=0.8	0.0116	0.0465	0.0249	0.0163	0.0412	0.0162
M4 Ens w=0.95 (ML)	0.0133	0.0481	0.0207	0.0176	0.0460	0.0176
M0 Direct SSR	0.0146	0.0474	0.0197	0.0176	0.0487	0.0183

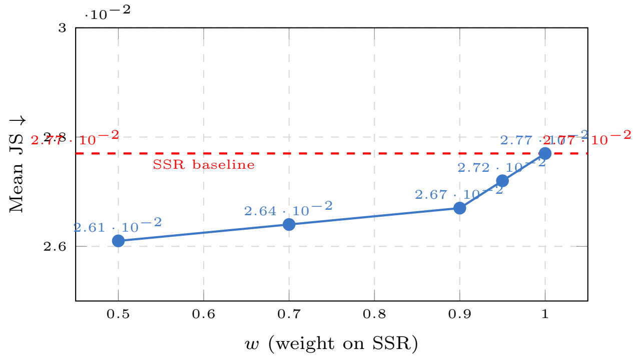
Note: M2 wins Q1, Q4, Q5, Q6; loses Q2, Q3. M1 (LLM-dist alone) extremely strong on Q1 & Q6.

4. M2 weight sweep (SSR + LLM-dist)



LLM-dist alone (w=0) nearly matches the best mix; SSR contributes a small correction at w=0.5.

5. M4 weight sweep (SSR + ML-steer)



Steering helps only as a minority voice; w=0.5 gives the largest independent gain.

6. Multi-layer steering alone (M3)

Layer set L	Mean JS
$[16, 24]$	0.0378
$[20, 24]$	0.0404
$[16, 20, 24, 28]$	0.0434
$[12, 16, 20, 24]$	0.0643
Single L24, α=0.1	0.0659

M3 alone never beats M0; it is useful only via the M4 ensemble.

7. M5 SSR knob tuning

Axis	Setting	Mean JS
Posts/cluster n	30	0.0277
	50	0.0275
	80	0.0276
Top-K clusters	5	0.0272
	10	0.0272
	15	0.0277
	20	0.0280

Effects within noise; tuning SSR alone cannot replace the ensemble.

8. Negative results (methods that do NOT beat pure SSR)

Method	Mean JS	vs M0
Per-post LLM voting (ask LLM per post)	0.2360	+755%
Large α=1.0 steering, unit-normed	0.0689	+149%
Single-layer steer L24, α=0.1	0.0659	+138%
Persona-projected SSR, s=0.1	0.0560	+102%
Persona-projected SSR, s=0.3	0.0478	+73%
Persona-projected SSR, s=0.5	0.0405	+46%
ICL with 5 example posts	0.0368	+33%
SSR + per-post LLM correction (not dist estimation)	0.0296	+7%
SSR + KL-penalty demographic reweight (from 23Q run)†	≈baseline	—
SSR + KL + QAW (from 23Q run)†	≈baseline	—

† Earlier 23Q-set claims; 23Q questions are LLM-generated synthetic items and do not reflect real reconstruction performance. Quarantined.

TL;DR. (1) **SSR + LLM-dist at** w=0.5 is the lead method: Mean JS = **0.0257**, a **7.2%** reduction over pure SSR. (2) The **LLM-dist** component carries most of the gain; SSR adds a small correction at 50/50 mix. (3) A second, **independent** contributor is the **SSR + multi-layer-steering ensemble** (M4), −5.4% at w=0.5; combining M2 + M4 is untested. (4) All single-layer steering, per-post LLM voting, persona-projection, and demographic KL/QAW methods are neutral or harmful.

Sources: `results/improve_orig6.json` (2026-04-21), `results/persona_methods.json` (2026-04-20), `results/fix_steering_results.json` (2026-04-19).